

and the corresponding Hamiltonian is

$$H(x, p, t) = p\dot{x} - L = \frac{1}{2m(t)}p^2 + \frac{1}{2}m(t)\omega^2(t)x^2. \quad (24)$$

Let us consider now a point transformation  $x \rightarrow \xi(x, t) = x/\rho(t)$ , which represents a time-dependent scaling transformation with the scaling function  $\rho(t)$  satisfying the auxiliary inhomogeneous equation

$$\ddot{\rho} + \gamma(t)\dot{\rho} + \omega^2(t)\rho = \frac{\Omega^2}{m^2(t)\rho^3}, \quad (25)$$

for  $\Omega$  constant. Without loss of generality, we will assume  $\rho(t) > 0$  [33]. Under this point transformation, we have:

$$L(x, \dot{x}, t) = L(\xi, \dot{\xi}, t) + \frac{d}{dt} \left( \frac{1}{2}m(t)\rho\dot{\rho}\xi^2 \right), \quad (26)$$

where

$$\begin{aligned} L(\xi, \dot{\xi}, t) &= \frac{m(t)}{2} \left[ \rho^2 \dot{\xi}^2 - \rho(\ddot{\rho} + \gamma(t)\dot{\rho} + \omega^2(t)\rho)\xi^2 \right] \\ &= \frac{m(t)}{2} \left[ \rho^2 \dot{\xi}^2 - \left( \frac{\Omega^2}{m^2(t)\rho^2} \right) \xi^2 \right], \end{aligned} \quad (27)$$

where the last line was obtained in view of (25). As it is well-known, the total derivative term in (26) does not affect the classical equation of motion; however, it plays a fundamental role as a generating function for canonical transformations. Further, in quantum mechanics, such a total derivative term manifests as a nontrivial time-dependent phase of the wave function [34]. Now, by transforming the time variable  $t \rightarrow \tau(t)$ , the Lagrangian (26) is mapped to

$$L\left(\xi, \frac{d\xi}{d\tau}, \tau\right) = \frac{m(\tau)}{2\dot{\tau}} \left[ \rho^2 \dot{\tau}^2 \left( \frac{d\xi}{d\tau} \right)^2 - \left( \frac{\Omega^2}{m^2(\tau)\rho^2} \right) \xi^2 \right], \quad (28)$$

where the global parameter  $\dot{\tau}$  in the denominator emerges to compensate the change of measure in the action induced by the transformation  $t \rightarrow \tau$ , and both  $\rho$  and  $\xi$  in this expression must be thought of as functions of the parameter  $\tau$ . In general, the new time variable  $\tau(t)$  may be considered as a nonlinear function in  $t$ . In particular, we can redefine the time variable  $\tau(t)$  such that

$$\dot{\tau} = \frac{1}{m(t)\rho^2(t)}. \quad (29)$$

This transformation implies that the coefficient of the kinetic energy term in equation (28) becomes constant. Consequently, we obtain

$$L\left(\xi, \frac{d\xi}{d\tau}, \tau\right) = \frac{1}{2} \left( \left( \frac{d\xi}{d\tau} \right)^2 - \Omega^2 \xi^2 \right). \quad (30)$$

This Lagrangian describes the motion of a simple harmonic oscillator with constant frequency  $\Omega$ . The Euler-Lagrange equation for the dynamical variable  $\xi(\tau)$  is given by

$$\frac{d^2\xi}{d\tau^2} + \Omega^2\xi = 0. \quad (31)$$